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Winter semester 2024/25

Exam Advanced Monetary Economics, 2nd Term
The exam time is 60 minutes. There are 60 points (P) overall. All questions are compulsory and need to be answered alone and in English. Cooperations are not allowed and will be valued as failure.

1. **[20 P] ECB policy**

   (a) **[6 P]** Name and explain the two types of climate risk that the ECB distinguishes. Give one example for each risk.

   (b) **[4 P]** What is the primary objective of the ECB? Explain the difference to the primary objective of the Federal Reserve.

   (c) **[4 P]** Name and explain two ways how the ECB tries to reduce the climate risks to the financial sector.

   (d) **[6 P]** Suppose that a flood destroys some of the technology in the economy and lowers productivity. What are the effects on prices and output according to the New Keynesian model? How would the ECB respond with its interest rate?

2. **[15 P] Monetary policy shocks**

   (a) **[4 P]** Explain the concept of the Taylor-rule.

   (b) **[4 P]** What is a monetary policy shock in the New Keynesian model? What are the effects of an expansionary monetary policy shock on output and inflation in the model?

   (c) **[7 P]** What is a conventional expansionary monetary policy shock in the market for reserves? Show it graphically.

3. **[25 P] Money-in-the-utility**

   Suppose that the representative household lives forever and has the instantaneous utility function

$$u\left(c_t, \frac{M_t}{P_t}\right) = ln(c_t) + \frac{D}{1-\gamma}\left(\frac{M_t}{P_t}\right)^{(1-\gamma)},$$

   which depends positively on real consumption c_t, $\frac{\partial u}{\partial c} > 0$ and real money holdings $m_t \equiv \frac{M_t}{P_t}$, $\frac{\partial u}{\partial m} > 0$. Future utility is discounted with the factor β. The household consumes $P_t c_t$ and it can invest in nominal money M_t and in nominal bonds B_t. Nominal bonds pay the gross interest rate $1 + i_{t-1}$. P_t is the aggregate price level. D and γ are preference parameters.

(a) **[4 P]** Setup the nominal budget constraint of the household. Derive the real budget constraint using the definition $\pi_t = P_t/P_{t-1}$.

(b) **[10 P]** Set up the Lagrangian and derive the first order conditions.

(c) **[6 P]** Provide an economic interpretation of each first order condition.

(d) **[5 P]** Derive $\frac{\partial u \partial u}{\partial c \partial m} = u_{cm}$. What does it mean (in words) and what does it imply for the effects of monetary policy shocks on nominal and real variables in the model?